

Taxonomy of detection algorithms for hyperspectral imaging applications

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Abstract. A unified, simplified, and concise overview of spectral target detection algorithms for hyperspectral imaging applications is presented. We focus on detection algorithms derived using established statistical techniques and whose performance is predictable under reasonable assumptions about hyperspectral imaging data. The emphasis on a signal processing perspective enables us to better understand the strengths and limitations of each algorithm, avoid unrealistic performance expectations, and apply an algorithm properly and sensibly. © 2005 Society of Photo-Optical Instrumentation Engineers. [DOI: 10.1117/1.1930927]

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1 Introduction

The detection of materials and objects using remotely sensed spectral information has many military and civilian applications. Hyperspectral imaging sensors measure the radiance for every pixel at a large number (K) of narrow spectral bands. The obtained measurements, arranged as a column vector $\mathbf{x}$, are known as the radiance spectrum of the pixel. In the reflective part of the electromagnetic spectrum (0.4 to 2.5 μm), the spectral information characterizing a material is the reflectance spectrum, defined as the ratio between reflected and incident radiation as a function of wavelength. In the emissive part of the spectrum (8 to 14 μm), the quantity characterizing a material is its emissivity spectrum, which describes how well an object radiates energy compared to the perfect blackbody radiator. Due to the effects of illumination, atmospheric transmission, and sensor characteristics, the radiance spectrum will be different than the reflectance or emissivity spectrum of the pixel. The estimation of the reflectance or emissivity spectrum from the radiance spectrum (atmospheric compensation) is a crucial step in most hyperspectral imaging exploitation applications. The algorithms presented in this paper can be used with radiance, reflectance, or emissivity measurements as long as all involved quantities are of the same kind.

To detect a target based on its spatial properties (for example, size, shape, texture), the target must be large relative to the pixel size. In this case, some of the pixels are fully occupied by one or more target materials (full pixel or resolved targets). If the spectrum of these target pixels is distinct from the surrounding background (that is, nontarget) pixels, the target can be detected spectrally as well as spatially. If a target occupies only part of a pixel (subpixel or unresolved target), it cannot be detected using spatial techniques. However, under certain conditions, it is possible to detect subpixel targets spectrally. The spectrum of a subpixel target is mixed with the spectrum (or spectra) of the background, which results in a combined spectrum. The

detectability of a full-pixel target depends on the spectral contrast between target and background. The detectability of subpixel targets is also dependent on the portion of the pixel occupied by the target (fill factor), because it determines the amount of background interference on the observed target spectrum. Additional factors affecting spectral detectability include environmental conditions, sensor noise, and statistical variability of target and background spectra.

In this paper, we focus on detection algorithms that exploit spectral information, only. Target detection algorithms for hyperspectral imaging data, can be grouped into two types:

1. *Spectral matching detection algorithms* that require spectral information about the targets of interest, and they try to identify pixels whose spectrum exhibits a high degree of correlation (“matches”) to the expected signature. The spectrum of the target of interest (reference spectrum) can be obtained from a spectral library or from an identified in-scene target pixel.
2. *Spectral anomaly detection algorithms* that do not require knowledge of the spectral signatures of the targets of interest. Every pixel whose spectrum does not fit a model of the local or global background is declared to be a target. Anomaly detectors cannot distinguish among legitimate spectral anomalies generated by manmade objects, natural objects, or targets of interest. Clearly, atmospheric compensation is not a prerequisite for the application of anomaly detection algorithms.

There are two key observations regarding spectral target detection algorithms. First, for any target detection algorithm to be successful, the target spectrum should be distinguishable from background spectra. Second, spectral matching algorithms exhibit better detection performance than anomaly detectors because they look for specific spectrally defined targets. Clearly, anomaly detectors are useful

for target-cueing applications and may perform satisfactorily for isolated manmade targets embedded in natural homogeneous backgrounds.

The main objective of this paper is to provide a concise and systematic review of some key algorithms for hyperspectral target detection and explain several issues regarding their practical application. This discussion expands and complements the work presented in Refs. 1 and 2. The paper is organized as follows. Section 2 introduces the basic ideas for detector design and evaluation using the likelihood ratio test approach. In Sec. 3, we introduce parametric probability models for target and background spectra. We stress the relationship between normal models and subspace normal models. These models provide the basis for detector design in Sec. 4 (full-pixel target detection) and Sec. 5 (subpixel target detection). It is shown that structurally similar algorithms can be derived using different approaches and assumptions. This is exploited in Sec. 2 to obtain a taxonomy of the structurally different algorithms, which leads to a set of fundamental detection algorithm kernels. This section concludes with a discussion of some practical issues related to the use of the different algorithmic kernels in actual applications.

2 Design and Performance Evaluation of Detection Algorithms

The task of a target detection algorithm is to decide, by means of a statistical hypothesis test, whether a target of interest is present or not present in a pixel under test (PUT), based on the observed spectrum $\mathbf{x}$. We typically use a binary hypothesis test to choose between the following competing null and alternative hypotheses:

H_0 : Target absent,

H_1 : Target present.

According to the Neyman-Pearson (NP) criterion,³ the optimum decision strategy (maximize the probability of detection P_D while keep the probability of false alarm P_{FA} under a certain value) is given by the likelihood ratio test (LRT)

$$\Lambda(\mathbf{x}) = \frac{f_1(\mathbf{x}|H_1)}{f_0(\mathbf{x}|H_0)} \geq \eta, \quad (1)$$

where $f_0(\mathbf{x}|H_0)$ is the probability of observing $\mathbf{x}$ under the H_0 hypothesis, and $f_1(\mathbf{x}|H_1)$ is the probability of observing $\mathbf{x}$ under the H_1 hypothesis. The decision threshold η is set to yield the desired probability of false alarm P_{FA} .

To determine $\Lambda(\mathbf{x})$ it is necessary to know the conditional probability density functions (pdfs) $f_0(\mathbf{x}|H_0)$ and $f_1(\mathbf{x}|H_1)$. To derive practically useful algorithms, we introduce signal models that can be used to accurately express the required pdfs in terms of parameters that can be readily estimated from the available data or specified from *a priori* information.

Detector design requires obtaining explicit equations for the likelihood ratio $\Lambda(\mathbf{x})$. In contrast, detector performance evaluation is based on receiver operating characteristic

(ROC) curves. Therefore, there is a difference between obtaining performance and evaluating performance.

A ROC curve plots the P_D versus the P_{FA} as a function of the threshold η . Theoretical ROC curves can be obtained by making assumptions about the probability density functions under the two hypotheses. In practice, probabilities must be estimated from real data. Their accuracy depends on the number of available target and background pixels. Typically, a reliable estimate of a probability P requires at least $10/P$ pixels.⁴ Since the number of target pixels is often very limited, the estimate of P_D is not accurate. This implies that ROC curves should be used in empirical detection evaluation with extreme care. Additional considerations, such as constant false alarm rate (CFAR) operation and computational complexity, are important for practical applications.

3 Background and Target Signal Models

Under the H_0 hypothesis, the PUT consists only of background. Under the H_1 hypothesis, depending on the relative sizes of the target and the ground resolution cell, the PUT may consist of only target or target and background. When the PUT is filled with the target, we talk about resolved or full-pixel targets. When only part of the pixel is filled with the target, we talk about unresolved or subpixel targets. In this case, essentially, the target replaces part of the background in the PUT. If we denote by $\mathbf{s}$ the spectrum of the target, and by $\mathbf{v}$ the spectrum of the background, and by $\alpha \geq 0$ the fraction of the PUT area filled by the target, the observed spectrum is

$$\mathbf{x} = \alpha \mathbf{s} + (1 - \alpha) \mathbf{v}, \quad (2)$$

which is known as the replacement signal model for subpixel targets. The replacement model can be approximated by a linear signal model if $\alpha \ll 1$ ($\Rightarrow \mathbf{x} \approx \alpha \mathbf{s} + \mathbf{v}$) or $\alpha \approx 1$ [$\Rightarrow \mathbf{x} \approx \mathbf{s} + (1 - \alpha) \mathbf{v}$]. Although the replacement model makes better sense for subpixel targets, for mathematical simplicity we use the following linear model

$$\mathbf{x} = \alpha \mathbf{s} + \beta \mathbf{v}, \quad (3)$$

where α and β can take any values, positive or negative.

The target spectrum $\mathbf{s}$ and the background spectrum $\mathbf{v}$ can be modeled using a multivariate probability distribution or the linear mixing model (LMM):

$$\begin{aligned} \mathbf{x} &= \sum_{k=1}^M a_k \mathbf{s}_k + \mathbf{w} = [\mathbf{s}_1 \ \mathbf{s}_2 \ \dots \ \mathbf{s}_M] \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_M \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_M \end{bmatrix} \\ &= \mathbf{S} \mathbf{a} + \mathbf{w}, \end{aligned} \quad (4)$$

where $\mathbf{s}_k$ is the spectrum of each component (endmember), a_k is the fraction of each endmember (abundance), and $\mathbf{w}$ is the model fit error (part of the spectrum that is not modeled as a mixture of the endmembers or noise). Since the components of the spectral vectors are positive the abundance coefficients should satisfy the following constraints:

Table 1 Families of target detection algorithms.

Target Size	Target Model	Background Model	Detector Family	Section
Full pixel	$N(\boldsymbol{\mu}_s, \Gamma_s)$	$N(\boldsymbol{\mu}_b, \Gamma_b)$	Quadratic NP	4.1
	$N(\boldsymbol{\mu}_s, \Gamma)$	$N(\boldsymbol{\mu}_b, \Gamma)$	Matched filter	4.2
	Unknown	$N(\boldsymbol{\mu}_b, \Gamma)$	Anomaly detector	4.3
Subpixel	$\mathbf{S}=[\mathbf{s}_1 \mathbf{s}_2 \dots \mathbf{s}_P]$	$N(\boldsymbol{\mu}_b, \Gamma)$	GLRT covariance	5.1
	$\mathbf{S}=[\mathbf{s}_1 \mathbf{s}_2 \dots \mathbf{s}_P]$	$\mathbf{B}=[\mathbf{b}_1 \mathbf{b}_2 \dots \mathbf{b}_Q]$	GLRT subspace	5.2

$$a_k \geq 0 \quad \text{and} \quad a_1 + a_2 + \dots + a_M = 1. \quad (5)$$

However, these constraints are not always taken into consideration to ensure mathematical simplicity. If we relax the constraints, we have a linear subspace model.³ However, enforcing the constraints results to a convex hull model,⁵ which provides physically meaningful endmembers. We stress that the success of the LRT depends on how accurately we can determine the pdf of the PUT under the two hypotheses. The exact spectral composition of $\mathbf{s}$ or $\mathbf{v}$ is not directly relevant to the LRT [Eq. (1)].

To derive a target detection algorithm, we should decide on the type of the target (full or subpixel) and choose a model (probability distribution or linear subspace) for the target and the background spectra. This leads to different families of detection algorithms, which are summarized in Table 1. For reasons that become apparent later, we used the multivariate normal distribution [Eq. (6)] in the construction of this table. We note that the background parameters are estimated from the processed data cube, whereas the target parameters are provided by the user.

3.1 Statistical Background Modeling

The most widely used background model assumes that $\mathbf{v}$ is a random vector with a multivariate normal (or Gaussian) pdf given by

$$f(\mathbf{v}) = \frac{1}{(2\pi)^{K/2} |\Gamma|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{v} - \boldsymbol{\mu})^T \Gamma^{-1} (\mathbf{v} - \boldsymbol{\mu}) \right], \quad (6)$$

where $\boldsymbol{\mu} = E(\mathbf{v})$ is the mean vector, $\Gamma = E[(\mathbf{v} - \boldsymbol{\mu})(\mathbf{v} - \boldsymbol{\mu})^T]$ is the covariance matrix, and $|\cdot|$ denotes matrix determinant. In shorthand notation $\mathbf{v} \sim N(\boldsymbol{\mu}, \Gamma)$ or equivalently $g(\mathbf{v}; \boldsymbol{\mu}, \Gamma)$. The quadratic expression,

$$\Delta^2(\mathbf{v}; \boldsymbol{\mu}, \Gamma) = (\mathbf{v} - \boldsymbol{\mu})^T \Gamma^{-1} (\mathbf{v} - \boldsymbol{\mu}), \quad (7)$$

is a widely used statistical distance measure, known as the Mahalanobis distance. It can be shown that if $\mathbf{v} \sim N(\boldsymbol{\mu}, \Gamma)$, then $\Delta^2 \sim \chi_K^2$ (chi-squared distribution with K degrees of freedom).

The normal distribution model is widely used in classification and detection applications; however, it has been shown that it can accurately fit the body of the data distribution but not its tails.⁶ This is important in detection applications, because the tails of the distribution are responsible for the generation of false alarms. A more accurate statistical model for hyperspectral backgrounds, proposed

in Ref. 6 and investigated in Ref. 7, is the multivariate elliptically contoured (EC) t distribution given by

$$t_K(\mathbf{v}; \boldsymbol{\mu}, \mathbf{C}, M) = \frac{\Gamma[(K+M)/2]}{\Gamma(M/2) (\pi M)^{K/2} |\mathbf{C}|^{1/2}} \times \left[1 + \frac{1}{M} (\mathbf{v} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{v} - \boldsymbol{\mu}) \right]^{-K+M/2}, \quad (8)$$

where M is the number of degrees of freedom, $\mathbf{C} = (1 - 2/M)\Gamma$ is the scale matrix, and $\Gamma(\cdot)$ denotes the gamma function. The Mahalanobis distance is distributed as

$$\frac{1}{K} (\mathbf{v} - \boldsymbol{\mu})^T \mathbf{C}^{-1} (\mathbf{v} - \boldsymbol{\mu}) \sim F_{K,M}, \quad (9)$$

where $F_{K,M}$ is the F distribution with K and M degrees of freedom. The integer M controls the tails of the distribution: $M=1$ leads to the multivariate Cauchy distribution (heavier tails), whereas as $M \rightarrow \infty$, the EC t distribution approaches the multivariate normal distribution (lighter tails).

The empirical distribution of the Mahalanobis distance of a hyperspectral data cube, can be used to identify its joint spectral distribution by comparing to the theoretical chi-squared and F distributions corresponding to the multivariate normal and t distributions.

Airborne hyperspectral imagery data collected by the HYDICE sensor at the U.S. Army Aberdeen Proving Grounds, Maryland, on August 24, 1995, are used to illustrate the nature of the background distribution. HYDICE collects calibrated (postprocessed) spectral radiance data in 210 wavelengths spanning 0.4 to 2.5 μm in nominally 10-nm-wide bands. The image in Fig. 1 is a color representation of this data. Figure 2 illustrates that the estimated Mahalanobis distance of the HYDICE spectral data does not follow a chi-squared distribution. Instead, it has heavy tails which can be better modeled by F distributions. Therefore, the multidimensional spectra data are more accurately modeled using multivariate elliptical t distributions.

According to Yao's representation theorem,⁸ every random vector $\mathbf{v}$ with an EC distribution (ECD) can be expressed as a product

$$\mathbf{v} = \boldsymbol{\tau} \mathbf{z}, \quad (10)$$

Classes selected for statistical analysis		
Class name	Selection technique	Sample size
Grass	Spatially adjacent	7760
Trees	Spatially adjacent	8232
Mixed	Spatially adjacent	7590
Blocks 1–8	Spatially adjacent	25,000



Fig. 1 Color image representing the hyperspectral data from the HYDICE sensor data cube, showing the division of the data cube into rectangular blocks to reduce spatial inhomogeneity. These data are used to illustrate the empirical performance of different target detection algorithms. (Color online only.)

where τ is a nonnegative random variable with pdf $f_{\tau}(\tau)$ and $\mathbf{z} \sim N(0, \Gamma_s)$. If $\tau \sim 1/\sqrt{\chi_K^2}$, we obtain the t ECD in Eq. (8).

For a given value of τ , the pdf of $\mathbf{v}$ is given by $f(\mathbf{v}|\tau) = N(\mathbf{0}, \tau^2 \Gamma)$. The pdf of $\mathbf{v}$ is obtained by averaging over τ as follows:

$$f(\mathbf{v}) = \int_0^{\infty} g(\mathbf{x}; \mathbf{0}, \tau^2 \Gamma) f_{\tau}(\tau) d\tau. \quad (11)$$

Consequently, $\mathbf{v}$ is sometimes referred to as a compound Gaussian vector. The random variable τ represents the variations of the local background power in the PUT. In the

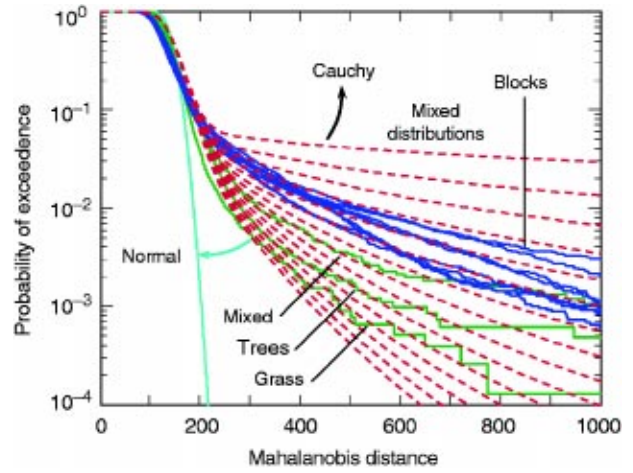


Fig. 2 Modeling the distribution of the Mahalanobis distance for the HSI data blocks shown in Fig. 1. The blue curves correspond to the eight equal-area subimages in Fig. 1. The green curves represent the smaller areas in Fig. 1 and correspond to trees, grass, and mixed (road and grass) materials. The dotted red curves represent the family of heavy-tailed distributions defined by Eq. (8). (Color online only.)

case of Gaussian backgrounds, $f_{\tau}(\tau) = \delta(\tau - \sigma^2)$, that is, the local power is no more random but $\tau = \sigma^2$ with probability one.

3.2 Subspace Background Modeling

The background subspace matrix $\mathbf{B}$ can be estimated from the data cube using statistical or geometrical techniques.⁹ To describe the ideas behind some of these methods, consider a matrix $\mathbf{X}^T$ of size $K \times N$, where each column contains the spectrum of an HSI pixel $\mathbf{x}(n)$, that is,

$$\mathbf{X}^T = [\mathbf{x}(1) \quad \mathbf{x}(2) \quad \dots \quad \mathbf{x}(N)] \quad (12)$$

where N is the total number of pixels. Background characterization for the detection of low-probability targets can be done using the eigenvectors^{10,11} of the HSI cube correlation matrix $\hat{\mathbf{R}}_x = \mathbf{X}^T \mathbf{X} / N$ or equivalently the singular vectors¹² of the data matrix $\mathbf{X}^T$. In the first case, matrix $\mathbf{B}$ is formed by the first Q significant eigenvectors of $\hat{\mathbf{R}}_x$. In the second case, $\mathbf{B}$ is formed by the first Q significant left-singular vectors of $\mathbf{X}^T$. The singular value decomposition (SVD) implies the following approximation:

$$\mathbf{X}^T = \mathbf{U} \Sigma \mathbf{V}^T = \sum_{k=1}^r \tilde{\sigma}_k \mathbf{u}_k \mathbf{v}_k^T \Rightarrow \hat{\mathbf{X}}^T = \sum_{k=1}^Q \tilde{\sigma}_k \mathbf{u}_k \mathbf{v}_k^T, \quad (13)$$

which is optimum in the sense of least squares error. It can be shown that

$$\sum_{n=1}^N \|\mathbf{x}(n) - \hat{\mathbf{x}}(n)\|^2 = \sum_{k=Q+1}^r \tilde{\sigma}_k^2, \quad (14)$$

where $\tilde{\sigma}_k$ are the singular values of $\mathbf{X}$. To obtain a good estimate of the background, the spectrum of interest should not be included among the significant eigenvectors or singular vectors. The columns of matrix $\mathbf{B}$ they do not corre-

spond to spectra of any physical material. This is not a problem for detection applications as long as $\mathbf{B}$ provides a good statistical approximation of the background and there is no leakage from the target subspace to the background subspace. The rarity of the target is a very important requirement in this respect; however, it is helpful to remove target-like pixels (that is pixels with large projections $\|\mathbf{S}^T \mathbf{x}(n)\|$ onto the target subspace) before computing the eigendecomposition or the SVD. Target leakage can be also reduced by excluding eigenvectors or singular vectors with large projections $\|\mathbf{S}^T \mathbf{u}_k\|$ onto the target subspace.¹² If we wish the elements of $\mathbf{B}$ to be actual endmembers, we should use one of the endmember extraction techniques available in the literature.¹³ In addition, using the method of independent component analysis (ICA) instead of principal component analysis (PCA) may provide spectra closer to physically observable spectra.

3.3 Target Modeling

In contrast to the background, the target spectrum $\mathbf{s}$ is usually modeled by a multivariate normal distribution:

$$\mathbf{s} \sim N(\boldsymbol{\mu}_s, \boldsymbol{\Gamma}_s). \quad (15)$$

Because the tails of the distribution do not significantly affect the probability of detection, there is no need to use the more complicated ECD models.

It can be shown that, the random signal model [Eq. (15)] is equivalent to the so-called Gaussian linear model

$$\mathbf{s} = \mathbf{H}\boldsymbol{\theta}, \quad \boldsymbol{\theta} \sim N(\boldsymbol{\mu}_\theta, \sigma_\theta^2 \mathbf{I}_P), \quad (16)$$

where $\mathbf{H}(K \times P)$ is the mode matrix and the elements of $\boldsymbol{\theta}(P \times 1)$ are the mode weights ($P \leq K$). According to this model, the target $\mathbf{s}$ lies in the linear subspace spanned by the columns of $\mathbf{H}$, but its location is unknown because $\boldsymbol{\theta}$ is unknown. If $\mathbf{H}$ and $\boldsymbol{\theta}$ satisfy the constraints of the linear mixing model [Eq. (5)], they can be interpreted as target signature spectra and their abundances.

To show the equivalence of Eqs. (15) and (16), we start with the eigenvalue decomposition

$$\boldsymbol{\Gamma}_s = \mathbf{U}_s \boldsymbol{\Lambda}_s \mathbf{U}_s^T, \quad (17)$$

where the columns of $\mathbf{U}_s$ consist of the P eigenvectors of $\boldsymbol{\Gamma}_s$, $\boldsymbol{\Lambda}_s(P \times P)$ contains the P nonzero eigenvalues, and $P = \text{rank}(\boldsymbol{\Gamma}_s) \leq K$.

The SVD of the rectangular matrix $\mathbf{H}$ is

$$\mathbf{H} = \mathbf{U}_H \boldsymbol{\Sigma}_H \mathbf{V}_H^T, \quad (18)$$

where $\mathbf{U}_H(K \times P)$ and $\mathbf{V}_H(K \times P)$ have orthogonal columns, and $\boldsymbol{\Sigma}_H(P \times P)$ contains the P nonzero singular values of $\mathbf{H}$.

Using Eqs. (20) and (23), the covariance matrix of $\mathbf{s}$ can be expressed as

$$\text{Cov}(\mathbf{s}) = \mathbf{H} \text{Cov}(\boldsymbol{\theta}) \mathbf{H}^T = \sigma_\theta^2 \mathbf{U}_H \boldsymbol{\Sigma}_H^2 \mathbf{U}_H^T. \quad (19)$$

Comparing Eq. (24) with Eq. (22), we see that the two models are equivalent if

$$\mathbf{U}_H = \mathbf{U}_s \quad (20)$$

$$\boldsymbol{\Lambda}_s = \sigma_\theta^2 \boldsymbol{\Sigma}_H^2. \quad (21)$$

Therefore, if $\text{rank}(\boldsymbol{\Gamma}_s) = P$, we can model the target spectrum either as a normal random vector [Eq. (15)] or as a subspace model

$$\mathbf{s} = \mathbf{U}_s \boldsymbol{\phi}, \quad \boldsymbol{\phi} \sim N(\boldsymbol{\mu}_\phi, \boldsymbol{\Lambda}_s), \quad (22)$$

where $\boldsymbol{\mu}_s = \mathbf{U}_s \boldsymbol{\mu}_\phi$. The projection matrix onto the column space of $\mathbf{s}$ can be equivalently be expressed as

$$\mathbf{P}_s = \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T = \mathbf{U}_H \mathbf{U}_H^T = \mathbf{U}_s \mathbf{U}_s^T. \quad (23)$$

in terms of $\mathbf{U}_s$. Also, we have $\text{rank}(\mathbf{P}_s) = \text{rank}(\mathbf{H}) = \text{rank}(\mathbf{U}_s) = P$.

The two statistically equivalent target models lead to different detection structures and interpretations. Furthermore, the subspace target model provides better understanding of the operation of the obtained detectors.*

4 Detection of Full-Pixel Targets

For full-pixel targets the competing hypotheses are

$$H_0: \mathbf{x} = \mathbf{v} = \tau \mathbf{z}, \quad (24)$$

$$H_1: \mathbf{x} = \mathbf{s}. \quad (25)$$

The computation of the NP-optimum LRT detector,

$$\Lambda(\mathbf{x}) = \frac{f(\mathbf{x}|H_1)}{f(\mathbf{x}|H_0)} = \frac{g(\mathbf{x}; \boldsymbol{\mu}_s, \boldsymbol{\Gamma}_s)}{\int_0^\infty g(\mathbf{x}; \boldsymbol{\mu}, \tau^2 \boldsymbol{\Gamma}) f_\tau(\tau) d\tau}, \quad (26)$$

requires the second-order moments and the pdf $f_\tau(\tau)$. The integration in the denominator of Eq. (26) makes the implementation of the optimum detector difficult. If a particular $f_\tau(\tau)$ leads to a parametric model for the ECD [see Eq. (8)], we could use this model to compute the LRT without integration.

4.1 Quadratic NP Detector

The computation of the LRT is simplified when the background is normally distributed. Indeed, it can be shown³ that the optimum-NP detector is given by

$$y = D(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}_b^{-1} (\mathbf{x} - \boldsymbol{\mu}_b) - (\mathbf{x} - \boldsymbol{\mu}_s)^T \boldsymbol{\Gamma}_s^{-1} (\mathbf{x} - \boldsymbol{\mu}_s), \quad (27)$$

which compares the Mahalanobis distance of the PUT from the means of background and target distributions.

4.2 Matched Filter Detector

If the background and target covariance matrices are equal, that is, $\boldsymbol{\Gamma}_s = \boldsymbol{\Gamma}_b \triangleq \boldsymbol{\Gamma}$, the optimum detector [Eq. (27)] becomes

*In radar applications, $P=1$ corresponds to Swerling I targets and $P=K$ to Swerling II targets.

$$y = D(\mathbf{x}) = \kappa(\boldsymbol{\mu}_s - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_b), \quad (28)$$

where κ is a normalization constant. The linear processor of Eq. (28) is known as the matched filter detector or Fisher's linear discriminant. The choice of κ does not affect the performance of the matched filter; in practice we set $\kappa = \Delta^{-2}(\boldsymbol{\mu}_s, \boldsymbol{\mu}_b; \boldsymbol{\Gamma})$ because $y = D(\boldsymbol{\mu}_s) = 1$. Thus, the NP-optimum matched filter (MF) detector is typically given by

$$y = D_{\text{MF}}(\mathbf{x}) = \frac{(\boldsymbol{\mu}_s - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_b)}{(\boldsymbol{\mu}_s - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}^{-1}(\boldsymbol{\mu}_s - \boldsymbol{\mu}_b)}. \quad (29)$$

To the degree that the normal model is accurate, the performance of the matched filter is completely determined by the Mahalanobis distance between the target and background distributions. The probability of detection and the probability of false alarm are related by

$$P_D = Q[Q^{-1}(P_{\text{FA}}) - \Delta], \quad (30)$$

where $\Delta = \Delta(\boldsymbol{\mu}_s, \boldsymbol{\mu}_b; \boldsymbol{\Gamma})$ and

$$Q(x) = \frac{1}{2\pi} \int_x^\infty \exp\left(-\frac{1}{2}t^2\right) dt \quad (31)$$

is the complementary cumulative distribution function.³

4.3 Anomaly Detection

In many situations of practical interest, we do not have sufficient *a priori* information to specify the statistics of the target class. Therefore, we can use neither the quadratic detector [Eq. (27)] nor the matched filter detector [Eq. (29)]. We clearly require a detector that exclusively uses background information. The result is the following, Mahalanobis-distance-based, LRT detector

$$y = D_{\text{MD}}(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}_b^{-1}(\mathbf{x} - \boldsymbol{\mu}_b), \quad (32)$$

which is NP-optimum when $\boldsymbol{\Gamma}_t = \boldsymbol{\Gamma}_s = \boldsymbol{\Gamma}$. The distribution of the detection statistics is

$$y = D(\mathbf{x}) \sim \begin{cases} \chi_K^2(0) & \text{under } H_0 \\ \chi_K^2(\Delta^2) & \text{under } H_1 \end{cases}, \quad (33)$$

where $\chi_K^2(\delta)$ is a noncentral chi-squared distribution with K degrees of freedom and noncentrality parameter δ . The detector of Eq. (32) is also known, in the remote sensing literature, as the RX detector.¹⁴

4.4 Detectability Index

In communications and radar systems, detection performance depends on the signal-to-noise ratio (SNR) or the signal-to-interference-plus-noise ratio (SINR). In hyperspectral imaging, background pixels and full target pixels are affected by the same kind of noise and interference (e.g., sensor noise, atmospheric noise). From Eqs. (30) and (33) we see that, for normally distributed target and background classes with equal covariances, detection performance is completely determined by the Mahalanobis distance between the two distributions:

$$\Delta^2 = (\boldsymbol{\mu}_s - \boldsymbol{\mu}_b)^T \boldsymbol{\Gamma}^{-1}(\boldsymbol{\mu}_s - \boldsymbol{\mu}_b), \quad (34)$$

which is known as the detectability index (DI). In this sense, the DI plays the role of SNR or SINR in hyperspectral target detection applications. However, SNR or SINR are more appropriate for additive noise and interference signal models.

4.5 Minimum Variance Distortionless Response and Constrained Energy Minimization Detectors

For the sake of completeness we provide a derivation of the matched filter algorithm and the constrained energy minimization algorithm using an optimization approach. Note, however, that the likelihood ratio approach used in this paper is the natural framework for target detection applications.

Given an observation $\mathbf{x}$ we form the linear combination

$$y = \mathbf{h}^T \mathbf{x}. \quad (35)$$

The variance of the random variable y is given by

$$\sigma_y^2 = \text{Var}(y) = \mathbf{h}^T E[(\mathbf{x} - \boldsymbol{\mu}_x)(\mathbf{x} - \boldsymbol{\mu}_x)^T] \mathbf{h} = \mathbf{h}^T \boldsymbol{\Gamma}_x \mathbf{h}. \quad (36)$$

The mean square value of the random variable y is given by

$$\mathcal{E} = E(y^2) = \mathbf{h}^T E(\mathbf{x}\mathbf{x}^T) \mathbf{h} = \mathbf{h}^T \mathbf{R}_x \mathbf{h}, \quad (37)$$

where $\mathbf{R}_x = E(\mathbf{x}\mathbf{x}^T)$ is the correlation (not covariance) matrix of the random vector $\mathbf{x}$.

Suppose that we wish to search for a target with a deterministic spectral signature $\mathbf{s}$. A reasonable approach would be to (1) choose $\mathbf{h}$ by minimizing the output variance or output mean square, subject to a constraint of unity response at the "direction" of the target spectrum; and (2) threshold the detector output y .

Solving the optimization problem

$$\min_{\mathbf{h}} \mathbf{h}^T \boldsymbol{\Gamma}_x \mathbf{h} \quad \text{subject to} \quad \mathbf{h}^T \mathbf{s} = 1 \quad (38)$$

leads to the following detector

$$y = D_{\text{MVDR}}(\mathbf{x}) = \frac{(\mathbf{s} - \boldsymbol{\mu}_x)^T \boldsymbol{\Gamma}_x^{-1}(\mathbf{x} - \boldsymbol{\mu}_x)}{(\mathbf{s} - \boldsymbol{\mu}_x)^T \boldsymbol{\Gamma}_x^{-1}(\mathbf{s} - \boldsymbol{\mu}_x)}. \quad (39)$$

The minimum variance distortionless response (MVDR) detector is similar, but not identical, to the matched filter of Eq. (29) because, in general, $\boldsymbol{\Gamma}_x \neq \boldsymbol{\Gamma}$. In practice, $\boldsymbol{\Gamma}_x$ is the covariance matrix of the entire cube, whereas $\boldsymbol{\Gamma}$ is the covariance matrix of the background.

In a similar fashion, solving the optimization problem

$$\min_{\mathbf{h}} \mathbf{h}^T \mathbf{R}_x \mathbf{h} \quad \text{subject to} \quad \mathbf{h}^T \mathbf{s} = 1 \quad (40)$$

leads to the following constrained energy minimization (CEM) algorithm¹⁵

$$y = D_{\text{CEM}}(\mathbf{x}) = \frac{\mathbf{s}^T \mathbf{R}_x^{-1} \mathbf{x}}{\mathbf{s}^T \mathbf{R}_x^{-1} \mathbf{s}}, \quad (41)$$

which is clearly different from Eq. (29) or Eq. (39). The CEM cannot be derived using a hypothesis test and, unlike the matched filter detector, it is not NP-optimum.

4.6 Practical Considerations

In practice, the required mean vectors and the background covariance or correlation matrices are unknown and should be replaced by estimates. The target mean is estimated by a signature $\hat{\boldsymbol{\mu}}_s$, which is determined using a single signature or the mean of a set of signatures from a spectral library. The mean and covariance matrix of the background are estimated from part or all the pixels in the analyzed data cube. The maximum likelihood estimates of the corresponding means, covariances, and correlation matrices are determined from a set of data $\{\mathbf{x}(n), 1 \leq n \leq N\}$ by

$$\hat{\boldsymbol{\mu}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}(n), \quad (42)$$

$$\hat{\boldsymbol{\Gamma}} = \frac{1}{N} \sum_{n=1}^N [\mathbf{x}(n) - \hat{\boldsymbol{\mu}}][\mathbf{x}(n) - \hat{\boldsymbol{\mu}}]^T, \quad (43)$$

$$\hat{\mathbf{R}} = \frac{1}{N} \sum_{n=1}^N \mathbf{x}(n)\mathbf{x}^T(n). \quad (44)$$

In some cases, we can do preliminary screening to remove “nonbackground” or “target-spoiled” pixels, before we estimate background statistics. Clearly, replacing exact moments with their estimates leads to deterioration of detection performance.

A more in-depth discussion of these algorithms can be found in Ref. 2. In practical applications, modeling the background detection statistics with an ECD distribution provides a better threshold selection strategy.

5 Detection of Subpixel Targets

The derivation of optimum-NP algorithms for full-pixel targets, assumed that the target spectra are normally distributed with a perfectly known mean spectrum. However, in most practical applications this spectrum is either known within a scaling factor or it is totally unknown. There are basically two approaches to deal with this problem: (1) to model the target spectrum by a signal that depends on unknown parameters and (2) to model the target spectrum as an unknown signal (deterministic or random) known to lie in a given subspace of rank P , $1 \leq P \leq K$ (see Sec. 4).

In this section, we adopt the subspace approach and we assume that the target subspace is perfectly known. This means that we know the subspace where each realization of the target spectrum is confined, but we do not know its direction. The spectrum vector can be placed deterministically or randomly in an unknown location of a known subspace. The rank of the subspace reflects the degree of uncertainty about the target and for randomly placed targets is given by the rank of target covariance matrix. Signals with known shape, that is known direction and unknown (deterministic or random) amplitude, have rank one. The uncertainty about the target is maximum for $P = K$.

5.1 Unstructured Background Models

Note that for subpixel targets only the mixing between target and background is relevant; mixing in background pixels is not relevant as long as we can accurately model their pdf. For subpixel targets, the competing hypotheses are[†]

$$\begin{aligned} H_0: \mathbf{x} &= \mathbf{v} \quad \text{Target absent,} \\ H_1: \mathbf{x} &= \mathbf{S}\mathbf{a} + \beta\mathbf{v} \quad \text{Target present.} \end{aligned} \quad (45)$$

Hence, $\mathbf{x} \sim N(\mathbf{0}, \boldsymbol{\Gamma})$ under H_0 and $\mathbf{x} \sim N(\mathbf{S}\mathbf{a}, \beta^2\boldsymbol{\Gamma})$ under H_1 . In addition, we assume that we have access to a set of training background pixels $\mathbf{x}(n)$, $1 \leq n \leq N$, that are independently and identically distributed. Practical ways to determine background training data are discussed in Sec. 6.2. The test pixel $\mathbf{x}$ and the training pixels are assumed statistically independent.

If we assume that $\beta=1$, the maximum likelihood estimator of $\mathbf{a}$ is given by

$$\hat{\mathbf{a}} = (\mathbf{S}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{S})^{-1} \mathbf{S}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{x}. \quad (46)$$

For $P=1$, we obtain the following simplified form

$$\hat{a} = \frac{\mathbf{s}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{x}}{\mathbf{s}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{s}}, \quad (47)$$

which is identical to the MVDR filter [Eq. (39)] when $\boldsymbol{\Gamma}_x = \boldsymbol{\Gamma}$.

Furthermore, using the generalized likelihood ratio approach, we obtain the following detector^{16,17}

$$D_K(\mathbf{x}) = \frac{\mathbf{x}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{S} (\mathbf{S}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{S})^{-1} \mathbf{S}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{x}}{N + \mathbf{x}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{x}} \underset{H_0}{\overset{H_1}{\geq}} \eta_K \quad (48)$$

where $\hat{\boldsymbol{\Gamma}}$ is the maximum likelihood estimate of the covariance matrix. Although there is no optimality test associated with the generalized likelihood ratio test (GLRT) approach,³ it leads to the design of useful practical detectors.

If $P=K$ and the matrix $\mathbf{S}$ has full rank (that is, $\mathbf{S}$ is invertible), Eq. (48) leads to the following detector¹⁸:

$$y = D_{MD}(\mathbf{x}) = \mathbf{x}^T \hat{\boldsymbol{\Gamma}}^{-1} \mathbf{x} \underset{H_0}{\overset{H_1}{\geq}} \eta_A, \quad (49)$$

which is identical to the anomaly detection algorithm for full pixel targets [Eq. (32)]. This anomaly detector was derived and introduced in the remote sensing literature, where it is known as the RX algorithm, by Reed and Yu¹⁴ using a multivariate analysis of variance formulation.

A key assumption in the derivation of Eq. (48) was that the covariance matrix of the background is the same under the two hypotheses. However, for subpixel targets the amount of background covered area is different under the two hypotheses. Therefore, it is more appropriate to assume

[†]Since HSI data have a nonzero mean, unless otherwise stated, we remove the estimated mean from both the data cube and the target subspace vectors to comply with this model.

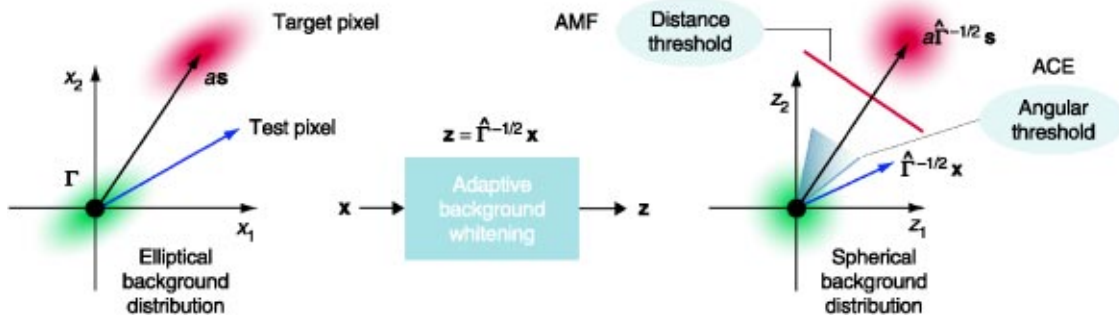


Fig. 3 Illustration of GLRT detectors. These are the AMF detector, which uses a distance threshold, and the ACE detector, which uses an angular threshold. The test-pixel vector and target-pixel vector are shown after the removal of the background mean vector.

β is unknown. In other words, the background has the same covariance structure but different variance. This variance is directly related to the fill factor of the target, that is, the percentage of the pixel area occupied by the target object. The GLRT approach leads to the following adaptive coherence/cosine estimator (ACE) detector^{19,20}:

$$D_{ACE}(\mathbf{x}) = \frac{\mathbf{x}^T \hat{\Gamma}^{-1} \mathbf{S} (\mathbf{S}^T \hat{\Gamma}^{-1} \mathbf{S})^{-1} \mathbf{S}^T \hat{\Gamma}^{-1} \mathbf{x} H_1}{\mathbf{x}^T \hat{\Gamma}^{-1} \mathbf{x}} \underset{H_0}{\geq} \eta_{ACE}, \quad (50)$$

which can be obtained from Eq. (48) by removing the additive term N from the denominator.

For targets with known shape (that is, for $P=1$), the target subspace $\mathbf{S}$ is specified by the direction of a single vector $\mathbf{s}$. Then, the formulas for the previous GLRT detectors are simplified to

$$y = D(\mathbf{x}) = \frac{(\mathbf{s}^T \hat{\Gamma} \mathbf{x})^2}{(\mathbf{s}^T \hat{\Gamma}^{-1} \mathbf{s})(\psi_1 + \psi_2 \mathbf{x}^T \hat{\Gamma}^{-1} \mathbf{x})} \underset{H_0}{\geq} \eta, \quad (51)$$

where $(\psi_1 = N$ and $\psi_2 = 1)$ for the Kelly detector and $(\psi_1 = 0$ and $\psi_2 = 1)$ for the ACE. Kelly's algorithm was derived for real-valued signals and was applied to multispectral target detection in Ref. 21. The 1-D version of ACE, has been derived in Refs. 22 and 23. Finally, we note that, if $(\psi_1 = N$ and $\psi_2 = 0)$, we obtain the adaptive matched filter (AMF) detector.^{24,25} Figure 3 provides a geometrical interpretation of the AMF and ACE algorithms.

It turns out that the distribution of the different detector outputs involves a noncentral F distribution.^{20,26} The noncentrality parameter is the theoretical SINR:

$$\text{SINR}_0 = (\mathbf{S}\mathbf{a})^T \hat{\Gamma}^{-1} (\mathbf{S}\mathbf{a}). \quad (52)$$

For subpixel targets the background acts as additive interference when the PUT contains target material. For this reason, it is more appropriate to use the term SINR instead of DI. However, the SINR in Eq. (52) is the Mahalanobis distance of the target vector $\mathbf{s}_t = \mathbf{S}\mathbf{a}$ from the zero-mean background.

The performance of all GLRT detectors depends only on the dimensional integer numbers K , P , and N and the optimum SINR_0 parameter. Under the H_0 hypothesis (target absent) $\text{SINR}_0 = 0$, the output distribution becomes central,

and the probability of false alarm depends only on the parameters K , P , and N . Therefore, all GLRT detectors discussed in this section have the CFAR property.

5.2 Structural Background Models

The subspace approach in Sec. 3.3 can be also used to model the spectral variability of the background. In this case, the detection problem involves choosing between the following competing hypotheses:

$$H_0: \mathbf{x} = \mathbf{B}\mathbf{a}_{b,0} + \mathbf{w} \quad (\text{Target absent}) \quad (53)$$

$$H_1: \mathbf{x} = \mathbf{S}\mathbf{a} + \mathbf{B}\mathbf{a}_{b,1} + \mathbf{w} \quad (\text{Target present}),$$

where the $(K \times P)$ matrix $\mathbf{S}$ must be specified by the user, the $(K \times Q)$ matrix $\mathbf{B}$ is determined from the data, and $\mathbf{w} \sim N(\mathbf{0}, \sigma_w^2 \mathbf{I})$.

Using the GLRT approach, leads to the following detection statistic^{3,27}:

$$y = D_{SB}(\mathbf{x}) = \frac{\mathbf{x}^T (\mathbf{P}_B^\perp - \mathbf{P}_{SB}^\perp) \mathbf{x}}{\mathbf{x}^T \mathbf{P}_{SB}^\perp \mathbf{x}} = \frac{(\text{TB})^2_{H_1}}{(\text{TC})^2_{H_0}} \underset{H_0}{\geq} \eta_{SB}, \quad (54)$$

where $\mathbf{P}_A^\perp = \mathbf{I} - \mathbf{P}_A$ and $\mathbf{P}_A = \mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ is the projection matrix onto the column space of matrix $\mathbf{A}$. The subscript SB denotes projection operators using the combined matrix $[\mathbf{S}\mathbf{B}]$. Figure 4 provides a geometrical interpretation of the GLRT detector for subspace modeled backgrounds.

The detection statistic is given by a noncentral F distribution with P numerator degrees of freedom, $K - P - Q$ denominator degrees of freedom, and noncentrality parameter SINR_0 :

$$\text{SINR}_0 = \frac{(\mathbf{S}\mathbf{a})^T \mathbf{P}_B^\perp (\mathbf{S}\mathbf{a})}{\sigma_w^2} = \frac{\|\mathbf{P}_B^\perp (\mathbf{S}\mathbf{a})\|^2}{\sigma_w^2}, \quad (55)$$

which depends on the length of the target projection onto the “background-free” subspace and the “everywhere-present” noise variance σ_w^2 . In the statistical literature $D_{SB}(\mathbf{x})$ is denoted by $F(\mathbf{x})$ and is known as the F test.

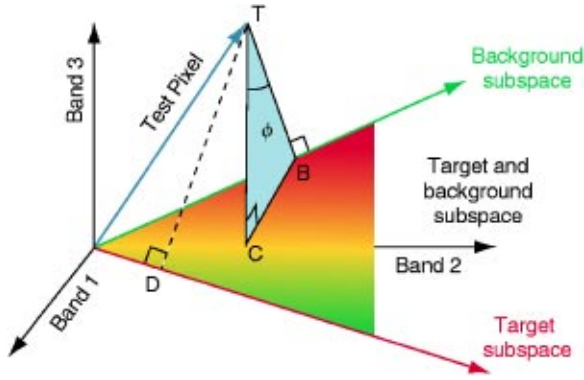


Fig. 4 Geometrical illustration of GLRT subspace target detector. The length of TC measures the goodness of fit for the full linear model (target plus background) to the test-pixel spectrum $\mathbf{x}$, whereas the length of TB measures the goodness of fit for the reduced model (background).

When $P=1$, it can be shown that the amplitude of the target signature in Eq. (53) can be estimated by the formula²⁸

$$\hat{a} = \frac{\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{x}}{\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{s}} \sim N[a, \sigma_w^2 (\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{s})^{-1}]. \quad (56)$$

Detection algorithms based on estimating $\hat{a}$ do not have the CFAR property, because σ_w^2 is unknown, and they are not optimum in the NP sense. The GLRT detector maximizes the SINR for any noise distribution and is CFAR in the case of Gaussian noise.

If we only take the numerator of Eq. (56), that is,

$$y = D_{\text{OSP}}(\mathbf{x}) = \mathbf{s}^T \mathbf{P}_B^\perp \mathbf{x}, \quad (57)$$

we obtain the detection algorithm known as orthogonal subspace projector²⁹ (OSP). Although the GLRT detector [Eq. (26)] and the OSP detector [Eq. (57)] use the same signal model [Eq. (53)], they have different properties and performance. Theoretically, the optimality and CFAR properties of GLRT algorithm are more desirable.

A slightly different normalization leads to the following “low-rank” subspace-ACE algorithm

$$y = D(\mathbf{x}) = \frac{(\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{x})^2}{(\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{s})(\mathbf{x}^T \mathbf{P}_B^\perp \mathbf{x})}. \quad (58)$$

5.3 Targets Placed Randomly in a Known Subspace

We now assume the target spectrum vector is randomly placed in a known subspace and we wish to know how this assumption affects the structure and performance of detection algorithms. We consider the following signal model:

$$\mathbf{x} = \mathbf{S}\mathbf{a} + \mathbf{v}, \quad (59)$$

where the subspace $\mathbf{S}$ and the covariance matrix $\mathbf{\Gamma}$ in $\mathbf{v} \sim N(\mathbf{0}, \mathbf{\Gamma})$ are assumed known. There are three cases of interest, leading to different algorithms.

5.3.1 Known gains $\mathbf{a}$

The target spectrum $\mathbf{s}_t = \mathbf{S}\mathbf{a}$ is perfectly known and the LRT is the matched filter

$$y = D(\mathbf{x}) = \mathbf{x}^T \mathbf{\Gamma}^{-1} (\mathbf{S}\mathbf{a}). \quad (60)$$

5.3.2 Unknown gains $\mathbf{a}$

The GLRT approach leads to the following detection statistic

$$y = D(\mathbf{x}) = \mathbf{x}^T \mathbf{\Gamma}^{-1} \mathbf{S} (\mathbf{S}^T \mathbf{\Gamma}^{-1} \mathbf{S})^{-1} \mathbf{S}^T \mathbf{\Gamma}^{-1} \mathbf{x}. \quad (61)$$

The whitening transformation $\tilde{\mathbf{x}} = \mathbf{\Gamma}^{-1/2} \mathbf{x}$ and $\tilde{\mathbf{S}} = \mathbf{\Gamma}^{-1/2} \mathbf{S}$, where $\mathbf{\Gamma}^{-1/2}$ is the Cholesky factor of $\mathbf{\Gamma}^{-1}$, gives the equivalent form

$$y = D(\mathbf{x}) = \|\mathbf{P}_{\tilde{\mathbf{S}}} \tilde{\mathbf{x}}\|^2, \quad (62)$$

which is the length of the component of $\mathbf{x}$ lying in the whitened target subspace.

5.3.3 Random gains $\mathbf{a}$

If $\mathbf{a} \sim N(\mathbf{0}, \mathbf{\Gamma}_a)$, the GLRT gives³

$$y = D(\mathbf{x}) = \sum_{k=1}^P \frac{\lambda_k}{1 + \lambda_k} (\mathbf{q}_k^T \mathbf{\Gamma}^{-1/2} \mathbf{x})^2, \quad (63)$$

where $\mathbf{q}_k$ are the eigenvectors corresponding to the dominant eigenvalues λ_k of the matrix $\mathbf{\Gamma}^{-1/2} \mathbf{S} \mathbf{\Gamma}_a \mathbf{S}^T \mathbf{\Gamma}^{-1/2}$. Basically the detector decides between the hypotheses $H_0: \mathbf{x} \sim N(\mathbf{0}, \mathbf{\Gamma})$ and $H_1: \mathbf{x} \sim N(\mathbf{0}, \mathbf{S} \mathbf{\Gamma}_a \mathbf{S}^T + \mathbf{\Gamma})$. A remarkable result developed in Ref. 30 shows that the subspace detectors for known covariance matrix are GLRT for both deterministic and stochastic subspace signals.

We note that the assumption about $\mathbf{a}$ (deterministic or random), although has the same practical implication (that is, unavailable $\mathbf{a}$), leads to different detector structures and performance. However, for one dimensional target subspaces, that is $\mathbf{S} = \mathbf{s}_1$, Eq. (63) becomes

$$y = D(\mathbf{x}) = \frac{(\mathbf{x}^T \mathbf{\Gamma}^{-1} \mathbf{s}_1)^2}{\mathbf{s}_1^T \mathbf{\Gamma}^{-1} \mathbf{s}_1}, \quad (64)$$

which is identical to Eq. (61) for $P=1$. We stress that although the two detectors have the same structure, their performance (that is, ROC curves and threshold selection) is different.

6 Algorithm Taxonomy and Practical Considerations

In Secs. 4 and 5 we have introduced detection algorithms based on various signal models. In this section, we provide a taxonomy of the fundamental detection algorithm kernels and discuss various issues regarding their practical implementation and application.

6.1 Algorithm Taxonomy

A careful examination shows that, in some cases, different signal models and approaches lead to algorithms described

Table 2 Fundamental detection algorithm kernels.

Name	Formula	Target (User)	Background (Automated)
Mahalanobis distance	$\mathbf{x}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{x}$	Unspecified	$\hat{\mathbf{\Gamma}}$
Matched filter	$\frac{\mathbf{s}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{x}}{\mathbf{s}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{s}}$	$\mathbf{s}$	$\hat{\mathbf{\Gamma}}$
CEM	$\frac{\mathbf{s}^T \hat{\mathbf{R}}_x^{-1} \mathbf{x}}{\mathbf{s}^T \hat{\mathbf{R}}_x^{-1} \mathbf{s}}$	$\mathbf{s}$	$\hat{\mathbf{R}}$
GLRT-AMF-ACE	$\frac{\mathbf{x}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{S} (\mathbf{S}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{S})^{-1} \hat{\mathbf{S}}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{x}}{\psi_1 + \psi_2 \mathbf{x}^T \hat{\mathbf{\Gamma}}^{-1} \mathbf{x}}$	$\mathbf{S}$	$\hat{\mathbf{\Gamma}}$
GLRT-SB	$\frac{\mathbf{x}^T \mathbf{P}_B \mathbf{x}}{\mathbf{x}^T \mathbf{P}_{SB}^\perp \mathbf{x}} - 1$	$\mathbf{S}$	$\mathbf{B}$
OSP	$\frac{\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{x}}{\mathbf{s}^T \mathbf{P}_B^\perp \mathbf{s}}$	$\mathbf{s}$	$\mathbf{B}$

by the same mathematical formula; only the meaning of some parameters may be different. For example, the matched filter [Eq. (29)], the MVDR filter [Eq. (39)], and the estimator [Eq. (47)] for subpixel targets have the same mathematical form. This observation can be used to consolidate the plethora of developed detectors, at least from an implementation point of view, into a compact set of basic algorithm kernels. The fundamental detection algorithm kernels discussed in this paper are summarized in Table 2. The interpretations, derivations, and theoretical properties of these algorithms provide the user with crucial guidance regarding the meaningful use of various detectors in practical applications.

6.2 Some Practical Considerations

The information-bearing signal in HSI applications, is the reflectance spectrum of each pixel in the data cube of the imaged area. The reflectance spectrum is a function of the physical properties and the structure of materials within the imaged pixel; in this sense, can be thought as a “spectral fingerprint” (signature) for these materials. However, the data cubes obtained by the HSI sensors consist of the radiance spectrum for each pixel. In principle, the radiance spectrum is obtained by “modifying” the solar spectrum by the reflectance spectrum of the pixel, the transmittance characteristics of the intervening atmosphere, and the response of the sensor. Besides bearing different information, radiance and reflectance spectra are measured in different units. Therefore, all data (that is, target signatures and data cubes) required by a detection algorithm should be either in the “reflectance-domain” or in the “radiance-domain.”

After we have made a choice between radiance-domain and reflectance-domain processing, we must determine the target and background parameters required by the detection algorithms in Table 2. The target signature $\mathbf{s}$ can be obtained from a spectral library. If several spectra, from a given target, are available, we have two choices.

1. Set $\mathbf{s}$ equal to the mean vector of the existing spectra.

2. Use an endmember selection techniques or the SVD to determine a low-dimensional target subspace $\mathbf{S}$.

The determination of the background subspace $\mathbf{B}$ is a challenging problem. It can be done by applying¹² SVD or endmember selection techniques to a set of background spectra from the data cube. Similarly, for detection algorithms that use the covariance matrix of the background, it is crucial how this matrix is estimated. It is extremely important that, the estimation of $\mathbf{b}$ and $\mathbf{\Gamma}$ should be done using a set of “target-free” pixels that accurately characterize the background. Several approaches to attain this objective include

1. Run a detection algorithm, remove a set of pixels that score high, recompute the covariance with the remaining pixels, and “rerun” the detection algorithm.
2. Before computing the covariance, remove the pixels with high projections onto the target subspace.¹²
3. Compute the covariance using a running window centered at the PUT. Pixels around the PUT are ignored using a guard mask to avoid spoiling the covariance matrix.
4. Use an unsupervised classification algorithm to segment the background into spectrally homogeneous classes and then perform detection per each class.

Another important consideration is the number of pixels required for a reliable and numerically invertible estimate of the covariance matrix. Certainly, the number of used pixels should exceed the number of bands; however, in practice, it should be much larger (typically, 10 to 100 times the number of bands) to obtain a statistically reliable estimate of the covariance matrix.

Finally, an important issue, which is often ignored in theoretical and empirical investigations, is the selection of the detection threshold η . In automated surveillance systems, the threshold should be chosen to assure an operationally useful CFAR. In a user-in-the-loop approach, inter-

active threshold selection through visual assessment may be useful. However, objective threshold selection requires accurate modeling of the background statistics. This topic is further discussed in Ref. 7.

7 Summary

In this paper, we presented a unified view of detection algorithms for hyperspectral imaging applications, from a signal processing perspective. We focused on algorithms that can be analytically derived and evaluated using statistical techniques. Various other variations or combinations of different algorithms are also available in the literature. Some of the discussed algorithms (GLRT, ACE, AMF) preserve their structure, which was obtained for normally distributed backgrounds, for the more general class of ECDs (Ref. 31). However, a definitive empirical performance evaluation of the different algorithms using real data is not yet available.

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